

GRIDBENCH · DESIGN PRECIS

GridBench design precis

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/02-concept.md @ 91bc975		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	TRL 3 update; design choices adopted as recommended (GBN-DDR-001); numbers from GBN-CAL-001; deeper clamp, diamond pin, split fence, ballast slabs; priced BOM
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, components, first-order numbers, safety, open questions)
0.1	2026-09-25	Amish Chadha	Initial scaffold

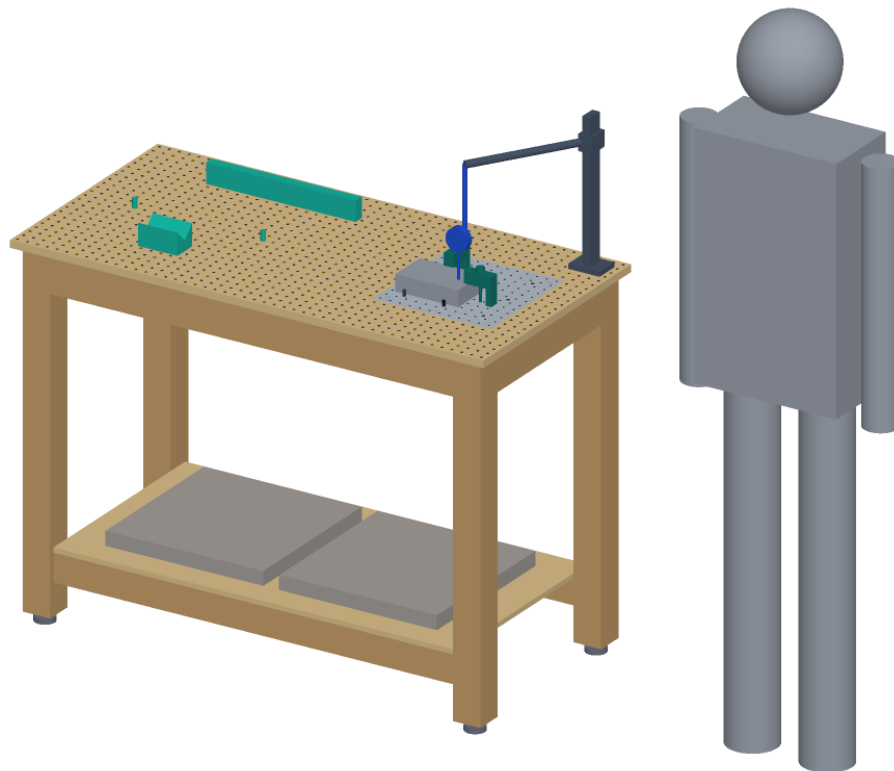
GridBench design precis

■ Summary

GridBench is a 1,200 x 600 mm workbench whose top is a 25 mm hole grid with M6 threads, the same pattern as metric optical breadboards. A cheap plywood field covers most of the top for general holding, and a 300 x 300 mm aluminum precision tile, set flush on the same grid, carries work that must repeat to a few hundredths of a millimeter. Printed clamps, stops, fences and V-blocks bolt to either zone, and a post with a user-supplied dial indicator checks parts in place. Two concrete slabs on the shelf keep the bench from tipping. The core parts cost \$239.20, above the \$220 budget; a \$250 fallback budget awaits Amish (see Cost). The parametric model is `cad/src/model.py`, the general arrangement is drawing GBN-DWG-001 and the calculations are GBN-CAL-001.

GridBench

CONCEPT, NOT FOR FABRICATION



Grey figure: 1.75 m person for scale

Figure 1. GridBench with a 1.75 m person for scale. Concept, not for fabrication.

■ How it works

1. **One grid everywhere.** Hole centers sit at $12.5 + 25k$ mm from the front-left corner of the worktop, in both directions. The tile's edges fall midway between grid lines, so its holes continue the plywood grid, a fixture can straddle both zones and any fixture file places itself by grid coordinates (column, row).

2. **Locate on dowels, clamp on threads.** As on machinist fixture plates, location and clamping are separate. On the tile, nine reamed 8 mm H7 bores at 100 mm pitch sit in the centers of grid squares, between the M6 holes. A precision fixture or part nest carries one round and one diamond (relieved) 8 mm h6 pin that drop into two bores, then M6 screws clamp it down. Two round pins would bind on any pitch error (GBN-CAL-001, section F). On the plywood field, 6.6 mm plain holes take stop pins, and M6 threaded inserts on a 50 mm sub-grid take clamps.
3. **Work and check in the same setup.** The instrument post bolts to any grid hole and carries a dial indicator on an arm, so a part can be checked before it is unclamped (Figure 3).
4. **Fixtures travel as files.** Each fixture is a parametric build123d part with a short metadata block that states its grid footprint and datum. A workshop that builds a GridBench can print or machine a published jig and expect it to fit. Jigs documented with the lab's ReadyKit toolchain can carry their own drawings and revision history.

GridBench: material flow for one part at a fixture (times are estimates)

CONCEPT, NOT FOR FABRICATION

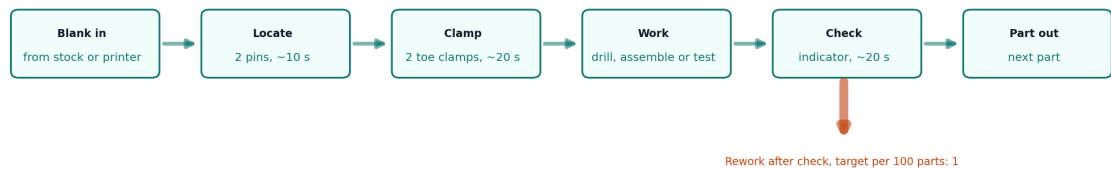


Figure 3. Material flow for one part at a fixture. Times and the rework rate are estimates or targets; the fixture change time is from GBN-CAL-001.

GridBench: exploded view

CONCEPT, NOT FOR FABRICATION

- 1 Bench frame, bolted timber
- 2 Levelling feet, M10 (4)
- 3 Lower shelf
- 14 Ballast slabs, concrete (2)
- 15 Wall anchor brackets (optional)
- 4 Plywood grid worktop (coarse field)
- 5 M6 threaded inserts, 50 mm sub-grid
- 6 Precision grid tile, aluminum
- 7 Round dowel pin, 8 mm h6
- 13 Diamond locating pin, 8 mm h6
- 8 Printed toe clamps
- 9 Printed fence, stops and V-block
- 10 Instrument post and arm
- 11 Dial indicator, 0.01 mm (user-supplied)

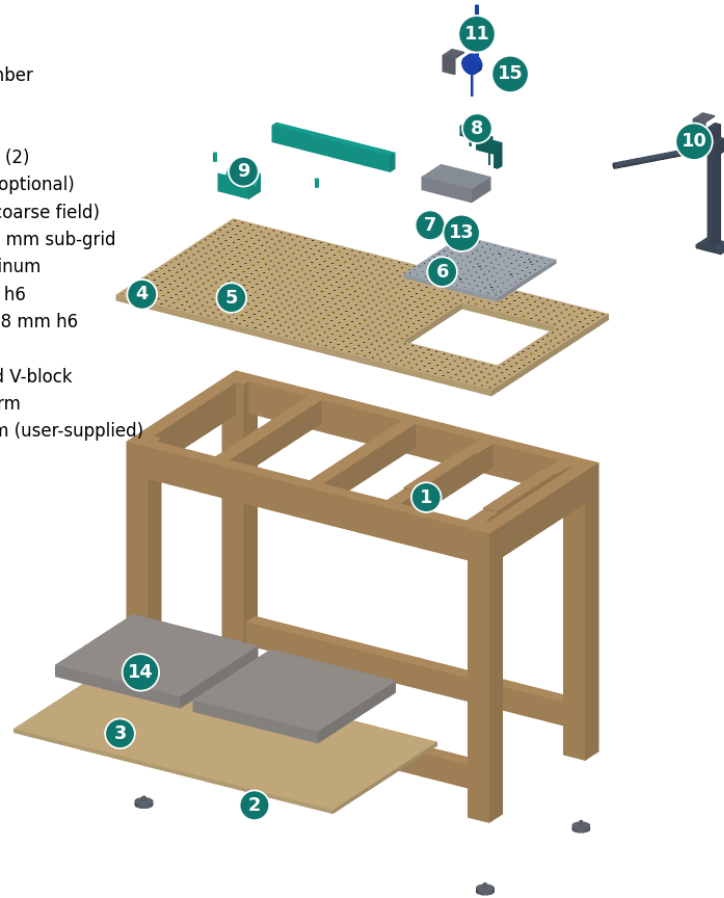


Figure 2. Exploded view. Callout numbers match the lines in `bom/bom.csv`.

■ Main components

Table 1. Main components (numbers match the BOM and Figure 2).

No.	Component	Design
1	Bench frame	Bolted softwood: four 70 x 70 mm legs at 1,090 x 490 mm centers, 45 x 95 mm aprons, four 45 x 70 mm cross rails (clear bays 255 mm or less), two low rails for the shelf, two 5.3 mm hardwood packers under the tile
2	Levelling feet	Four M10 feet in T-nuts, ± 15 mm adjustment
3	Lower shelf	12 mm plywood, 1,020 x 535 mm, resting on the low rails; carries the ballast
4	Plywood grid worktop	18 mm birch plywood, 1,200 x 600 mm, 1,008 holes on the 25 mm grid, 300 x 300 mm through pocket for the tile
5	M6 threaded inserts	252 screw-in inserts, 10 mm OD by 13 mm, on the 50 mm sub-grid
6	Precision grid tile	Cast aluminum tooling plate (offcut), 300 x 300 x 12.7 mm, 144 M6 tapped holes, 9 reamed 8 mm H7 bores, 4 counterbored fixing holes to the two cross rails
7	Round dowel pins	8 mm h6 hardened pins, 20 mm long; one per precision fixture
8	Printed toe clamps	PETG, 70 x 24 x 30 mm body with a 6.6 mm slot and a heel, printed on its side; rated screw torque 1.6 N·m
9	Printed fence, stops and V-block	Fence in two 190 mm segments, each on two grid holes; stop pins for 6.6 mm holes; 90° V-block
10	Instrument post and arm	20 x 40 mm aluminum extrusion on a printed grid foot, 18 mm arm
11	Dial indicator	0.01 mm resolution, 10 mm travel, 8 mm stem; user-supplied
12	M6 fastener kit	Cap screws, washers and printed knobs (no callout)
13	Diamond locating pins	8 mm h6 relieved pins; one per precision fixture, paired with a round pin
14	Ballast slabs	Two 400 x 400 x 35 mm concrete paving slabs, 12.9 kg each, on the shelf
15	Wall anchor brackets	Optional pair of steel angles from the rear apron to a wall
16	Tile machining tools	M6 taps, drills, 8 mm H7 reamer and a tapping guide (no callout)

■ Key numbers

All values are from GBN-CAL-001, where every assumption is stated. They are paper estimates, not test results.

Table 2. Key numbers.

Quantity	Value	Note
Grid positions	1,152 (48 x 24): 144 on the tile, 252 inserts and 756 plain holes	25 mm pitch, 12.5 mm edge offset
Bench mass	40.2 kg empty; 65.9 kg with two ballast slabs	Densities in GBN-CAL-001, Table 1
Tipping force at the top edge	107 N empty; 176 N with the ballast (target 150 N)	Feet 245 mm from the center line, push at 900 mm
Deflection, 500 N at bay center	0.19 mm in the plywood; 0.49 mm including rails and aprons (target 0.5 mm)	Plate theory, E about 7 GPa, pinned joints
Plywood field moisture movement	±0.115 mm over 1,000 mm at 40 to 60 % RH; ±0.40 mm over 20 to 80 % RH	Sourced moisture contents; plywood coefficient 0.007 % per % still unsourced
Tile thermal growth	0.069 mm over 300 mm per 10 K	Aluminum, 23×10^{-6} /K
Dowel relocation	0.036 mm worst case, 0.015 mm statistical (target 0.03 mm)	Round plus diamond pin, 200 mm apart
Clamp force at the part	515 to 858 N at 1.6 N·m; body stress 9.9 MPa or less	M6, nut factor 0.15 to 0.25
Insert pull-out	1.02 to 1.63 kN against up to 1.78 kN of screw tension	Assumed shear strength; R7 not met on the plywood field
Fixture change	56 s (target 60 s)	Two screws out, two in
Tile machining	About 4.2 h by drill press with a tapping guide	144 holes, 9 reamed bores, 4 counterbores

■ Key design choices

Each choice below is adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review (GBN-DDR-001).

1. **Grid: 25 mm, M6.** Matches metric optical breadboards, so commercial posts, clamps and bases fit (RP Photonics, "Optical breadboards"). A heavy 50 mm variant is left for later.
2. **Two zones instead of one precise top.** A full aluminum top of 1,200 x 600 mm would cost several times the budget. Plywood gives area; the tile gives precision where it is needed.
3. **Inserts on a 50 mm sub-grid,** with the 25 mm holes kept for pins. Inserts at every plywood hole would add about \$75 and a day of fitting.
4. **Bolted timber frame** for the prototype, with welded steel as a documented variant.
5. **PETG printed fixtures with steel where it matters.** Dowels, screws and the tile are metal; bodies are printed, with a machined option for heavily used jigs.
6. **Tile at the right of the worktop,** leaving the left 800 mm free for large parts.
7. **Ballast on the shelf plus an optional wall anchor,** for stability.
8. **Fixture files carry a metadata block** stating grid footprint and datum; the fixture library stays under CERN-OHL-S-2.0.

Four details changed at TRL 3 because of GBN-CAL-001, within these choices: the toe clamp body is 30 mm deep instead of 16 mm and its rated torque is 1.6 N·m instead of 2 N·m (the TRL 2 clamp would have carried about 33 MPa and crept); each precision fixture uses one round and one diamond pin; the fence is two 190 mm segments so that it fits a 200 mm print bed; and the ballast is two concrete slabs, 25.8 kg in all.

■ Cost

The priced BOM (`bom/bom.csv` , 16 lines) gives core parts of \$239.20, 8.7 % over the \$220 budget, with the dial indicator (\$15.00) user-supplied and the wall anchor brackets (\$5.00) optional; everything together is \$259.20. The largest items are the plywood worktop (\$45.00), the frame (\$41.40), the tile offcut (\$28.00) and the inserts (\$25.20). The cost options adopted in GBN-DDR-001 (indicator user-supplied, tile as offcut) are not enough on their own. Raising `budget_usd` to \$250 would cover the core parts with \$10.80 to spare; this is **Proposed, awaiting Amish**, and `project.yaml` is unchanged.

■ Safety

Safety: Drilling, tapping and routing the worktop and tile create chips and projectiles: wear eye protection and clamp work before cutting. Clamps and toe clamps create pinch points; keep fingers clear when tightening. The empty bench tips at about 107 N at the top edge (GBN-CAL-001), below the 150 N design push, so place both ballast slabs on the shelf or anchor the bench to a wall before use, and lift each 13 kg slab with a straight back. Chamfer or round all plywood, tile and printed edges. Printed clamps can crack and release a part under load; do not exceed the rated screw torque of 1.6 N·m, and do not use printed clamps for machining with a milling machine. Screw-in inserts in the plywood field may pull out before a clamp reaches its rated force. The bench is not for welding, grinding sparks or hot work: plywood and PETG are combustible.

■ Open questions

- [] R7 on the plywood field: tee nuts from below, a lower rated torque on the plywood, or a relaxed target. Proposed, awaiting Amish (`docs/REVIEW.md`).
- [] `budget_usd` of \$250 (GBN-DDR-001 O2). Proposed, awaiting Amish.
- [] First workshops and regions for co-design (GBN-DDR-001 O1). Proposed, awaiting Amish.
- [] Can the tile be drilled to ± 0.05 mm without CNC, using a printed or steel drill jig on a drill press? Not verifiable on paper.
- [] A sourced figure for plywood in-plane moisture movement; the moisture contents are now sourced, the coefficient is not.
- [] Flatness of cast tooling plate bought as offcut, and of the plywood sheet once screwed down (R5).